



# FEDERICO COLOMBO

Computer Graphics Engineer | Technical Artist | XR Developer

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## CORE SKILLS

**Programming:** C++, C#, Python

**Graphics & Shaders:** HLSL, GLSL, Compute Shaders, VFX Graph

**Engines & Tools:** Unity (Expert), Unreal Engine 5, Custom OpenGL Frameworks, Blender

**Workflows:** Git, Jira, Project Management, Agile Product Strategy

## EDUCATION

### M.Sc. Cinema and Media Engineering

Politecnico di Torino | 2024 - 2026

Expected mark: 110/110 Cum Laude (Dec 2026).

### B.Sc. Cinema and Media Engineering

Politecnico di Torino | 2021 - 2024

Final mark: 100/110.

## PERSONAL PROJECTS

### Smart Cable Gen (Blender Add-on)

Python / Pipeline Tooling

Developed a procedural multi-object cable authoring add-on in Python with scale-independent catenary physics and custom UI for game-ready export.

### Graphics Lab (Custom C++ Engine)

Engine Programmer

Engineered a custom OpenGL 3D rendering framework from scratch with Deferred Shading, PBR, and SPOM.

### UE5 Tech Art & Python Tools

Procedural Workflows & Rigging

Built procedural environments (PCG) and custom FK/IK character rigs. Developed Python scripts for pipeline automation and integrated MetaHumans.

### Awaken Asylum & Nihil

Lead Technical Artist

Developed an immersive VR educational experience (Unity) and a 2D Metroidvania prototype. Managed end-to-end technical production.

## PROFESSIONAL PROFILE

Computer Graphics Engineer and Technical Artist specialized in real-time rendering, spatial computing (XR), and procedural tool development. Graduated with honors in Cinema and Media Engineering. Deep expertise in writing custom C++ rendering engines, optimizing GPU architectures (HLSL/GLSL), and leading multidisciplinary teams. Actively seeking to leverage hardcore engineering and project management skills to build scalable software architectures and high-fidelity interactive productions.

## WORK EXPERIENCE

### XR Master Thesis - R&D Graphics Programmer 03/2026 - Present

#### LEMONS IN THE ROOM

Firenze, Italy

- Engineered a real-time HLSL Raymarching framework for Signed Distance Fields (SDFs) in Unity (URP), enabling procedural, infinite-resolution topological morphing driven by biometric data.
- Developed an in-engine authoring tool with custom Shader Graph nodes and dynamic parameter routing, allowing artists and clinicians to visually compose and map live BLE biometric streams (ECG, HRV, breathing) to SDF morphology.
- Implemented Dynamic Bounding Volumes and adaptive step-size scaling to cut GPU fill-rate costs, guaranteeing a stable 72Hz framerate on standalone Meta Quest 3 hardware.

### Technical Artist & Gameplay Programmer

01/2025 - Present

#### LEVEL UP LAB

Torino, Italy

- Programmed core gameplay mechanics in C# (Unity) and designed the technical architecture for the "Line Operator" project.
- Acted as Project Manager and Technical Producer, coordinating multidisciplinary teams of designers and artists to oversee artistic vision and technical viability.
- Developed custom HLSL Shaders and post-processing pipelines to establish the game's unique visual identity.

### Teaching Assistant (GUI)

10/2024 - 01/2025

#### POLITECNICO DI TORINO

Torino, Italy

- Tutored students in UI/UX principles and frontend programming during weekly laboratory sessions.
- Wrote technical documentation and created high-quality instructional content to improve comprehension.

### Video Artist

09/2023 - 06/2024

#### RKH SRL

Torino, Italy

- Operated professional cameras (RED, Blackmagic 4K/6K) and assisted DOP with complex lighting setups.
- Handled editing and post-production workflows using DaVinci Resolve for corporate content.